

A Local Counterexample to Operator-Level Entropy Extremality for Stokes' Theorem

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Source assertion

Daniel Lazarev, *Stokes' theorem as an entropy-extremizing duality*, arXiv:2509.16386v2, closes by considering alternative derivative operators $\mathfrak{d}$ satisfying the Stokes integral relation and reports preliminary evidence that entropy is uniquely maximized at $\mathfrak{d} = \mathrm{d}$.

STOKES' THEOREM AS AN ENTROPY-EXTREMIZING DUALITY

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globally satisfy the Stokes relation: Given $\omega \in \Omega^{n-1}(\mathcal{M})$, one may define the alternative derivative operators $\mathfrak{d} : \Omega^{n-1}(\mathcal{M}) \rightarrow \Omega^n(\mathcal{M})$ such that

$$\int_{\mathcal{M}} \mathfrak{d}\omega = \int_{\partial\mathcal{M}} \omega ,$$

that is, operators indistinguishable from the exterior derivative, d , when applied to ω and integrated over $\mathcal{M}$. Preliminary investigations indicate that among all such $\mathfrak{d}$, entropy is similarly uniquely maximized for $\mathfrak{d} = \mathrm{d}$.

These observations point toward a more comprehensive theory of entropy-extremizing dualities that could encompass classical dualities from convex analysis, Hodge theory, and functional inequalities, and that may, more importantly, provide a rigorous variational framework for the principled discovery of new results. A systematic development of this framework is the focus of future work.

Precise interpretation and verdict

The source defines $S^*(\alpha, M)$ for a top form α , but does not separately define entropy of an operator. The direct extension is therefore to compare

$$S^*(\mathfrak{d}\omega, M).$$

Under this natural formulation the assertion is false, even for smooth local first-order linear operators satisfying Stokes' identity for *every* smooth input. If a different operator entropy is intended, it must first be defined.

Theorem 1 (Smooth local counterexample). *Let $M = [0, 1]$ with its standard orientation. For every $0 < \varepsilon < 1$ there is a local first-order linear operator*

$$\mathfrak{d}_\varepsilon : C^\infty([0, 1]) \longrightarrow \Omega^1([0, 1])$$

such that

$$\int_M \mathfrak{d}_\varepsilon g = g(1) - g(0) = \int_{\partial M} g \quad \text{for every } g \in C^\infty([0, 1]), \tag{1}$$

but, for $\omega(x) = x$,

$$S^*(\mathfrak{d}_\varepsilon \omega, M) > S^*(\mathrm{d}\omega, M) = 0. \tag{2}$$

Consequently the exterior derivative is not an entropy maximizer in the announced competitor class.

Proof. For $x > 0$ set

$$a_\varepsilon(x) = 1 + \varepsilon \frac{1 - \cos(2\pi x)}{2\pi x}, \quad a_\varepsilon(0) = 1,$$

and define

$$\mathfrak{d}_\varepsilon g = \mathrm{d}(a_\varepsilon g) = (a_\varepsilon g' + a'_\varepsilon g) \, \mathrm{d}x. \quad (3)$$

The Taylor series of $1 - \cos(2\pi x)$ shows that a_ε extends smoothly at zero. Moreover $a_\varepsilon(0) = a_\varepsilon(1) = 1$. The fundamental theorem of calculus now gives (1), so this is a globally Stokes-compatible operator, not merely one tailored to a single ω .

For $\omega(x) = x$, direct differentiation gives

$$\mathfrak{d}_\varepsilon \omega = \mathrm{d}\left(x + \varepsilon \frac{1 - \cos(2\pi x)}{2\pi}\right) = (1 + \varepsilon \sin(2\pi x)) \, \mathrm{d}x. \quad (4)$$

Its density is $\rho_\varepsilon(x) = 1 + \varepsilon \sin(2\pi x) > 0$ and its source normalization is $C = \|\rho_\varepsilon\|_\infty = 1 + \varepsilon$. Hence

$$S^*(\mathfrak{d}_\varepsilon \omega, M) = \int_0^1 \rho_\varepsilon(x) \log \frac{1 + \varepsilon}{\rho_\varepsilon(x)} \, \mathrm{d}x > 0. \quad (5)$$

Indeed the integrand is nonnegative everywhere and positive except on the measure-zero set where $\sin(2\pi x) = 1$. On the other hand $\mathrm{d}\omega = \mathrm{d}x$ has constant density one and $C = 1$, so its entropy is zero. This proves (2). Notice that nonnegativity of S^* makes d a minimum, not a maximum, for this example. $\square$

What extra axioms can rescue the claim?

The construction (3) is already local and first order. Thus locality alone does not remove the obstruction. The natural stronger axiom is the Leibniz rule.

Proposition 1 (Derivation rigidity). *Suppose a scalar local first-order operator on $[0, 1]$ is a derivation and satisfies (1) for every smooth input. Then it equals d .*

Proof. Every such derivation has the form $Dg = a(x)g'(x) \, \mathrm{d}x$. For a compactly supported test function g , (1) gives $\int_0^1 ag' = 0$; distributional integration by parts yields $a' = 0$. Thus a is constant. Taking $g(x) = x$ in (1) then gives $a = 1$. $\square$

This produces a sharp dichotomy. In the broad linear or local first-order class announced in the source, Theorem 1 disproves entropy maximality. In the derivation class, Stokes compatibility already forces the sole competitor to be d , so uniqueness is algebraic and the entropy variational principle is vacuous.

Verification and scope

The accompanying script symbolically checks the endpoint values of a_ε , (4), and the Stokes integral, and numerically evaluates (5) at $\varepsilon = 1/2$ with 50-digit precision. A bounded search of the run indexes, exact source wording, arXiv, and current web results found no later formulation or proof of the operator assertion. Specialist review should focus on whether the author intended additional competitor axioms or a different entropy functional.